

torch.autograd.Function.vmap#

<https://pytorch.org/docs/stable/generated/torch.autograd.Function.vmap.html>

Description:

Define the behavior for this `autograd.Function` underneath `torch.vmap()`. For a `torch.autograd.Function()` to support `torch.vmap()`, you must either override this static method, or set `generate_vmap_rule` to `True` (you may not do both). If you choose to override this static method: it must accept an `info` object as the first argument. `info.batch_size` specifies the size of the dimension being vmapped over, while `info.randomness` is the randomness option passed to `torch.vmap()`. an `in_dims` tuple as the second argument. For each `arg` in `args`, `in_dims` has a corresponding `Optional[int]`. It is `None` if the `arg` is not a `Tensor` or if the `arg` is not being vmapped over, otherwise, it is an integer specifying what dimension of the `Tensor` is being vmapped over.

Function Signature

```
static Function.vmap(info, in_dims, *args)[source]#
```

Detailed Documentation

Documentation

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For a `torch.autograd.Function()` to support `torch.vmap()`, you must either override this static method, or set `generate_vmap_rule` to `True` (you may not do both).

If you choose to override this staticmethod: it must accept

an `info` object as the first argument. `info.batch_size` specifies the size of the dimension being vmapped over, while `info.randomness` is the randomness option passed to `torch.vmap()`.

an `in_dims` tuple as the second argument. For each `arg` in `args`, `in_dims` has a corresponding `Optional[int]`. It is `None` if the `arg` is not a `Tensor` or if the `arg` is not being vmapped over, otherwise, it is an integer specifying what dimension of the `Tensor` is being vmapped over.

`*args`, which is the same as the `args` to `forward()`.

The return of the `vmap` staticmethod is a tuple of `(output, out_dims)`. Similar to `in_dims`, `out_dims` should be of the same structure as `output` and contain one `out_dim` per output that specifies if the output has the vmapped dimension and what index it is in.

Please see [Extending torch.func with autograd.Function](#) for more details.